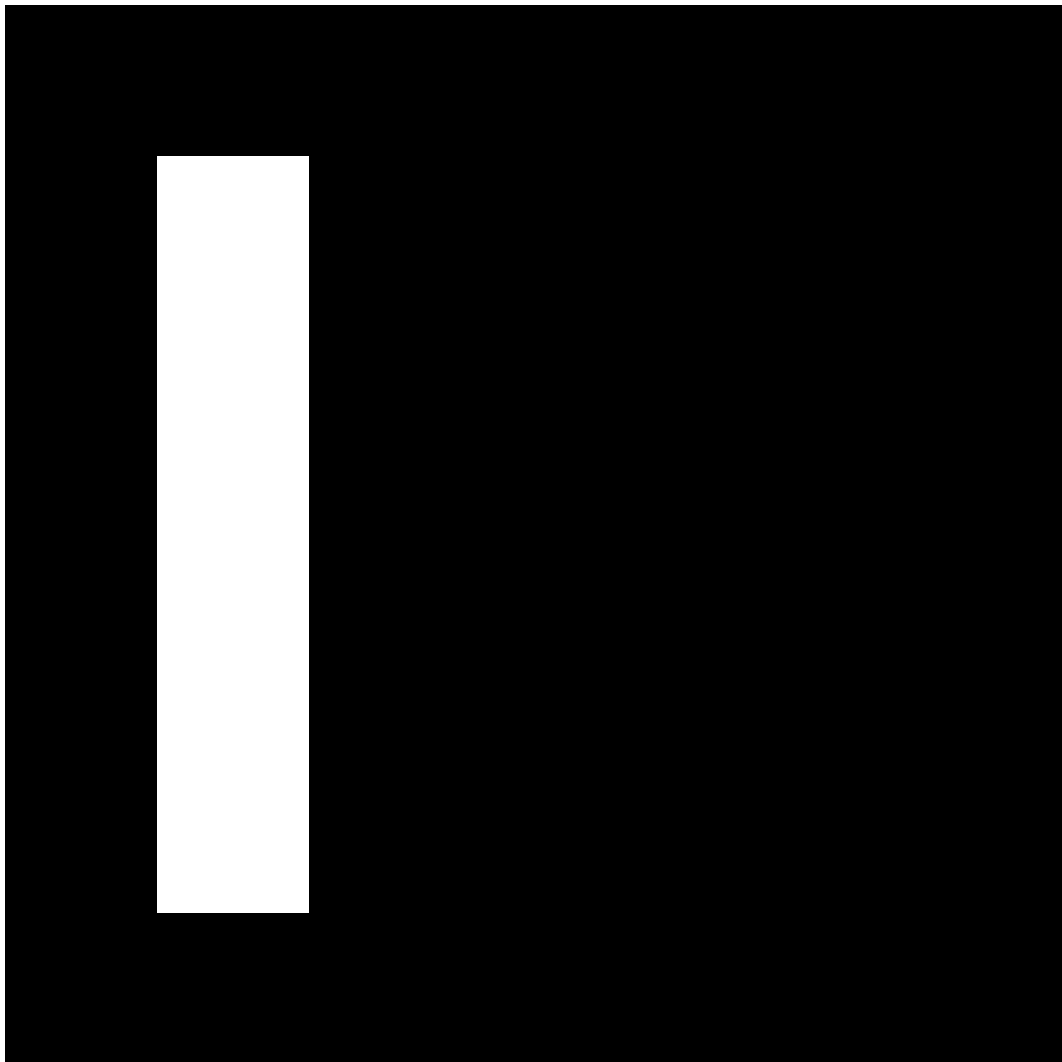
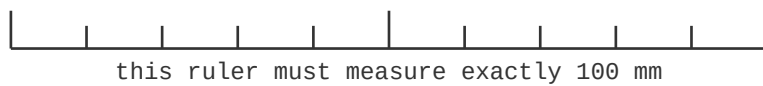




ars // ARUCO id 0

black square = 140 mm -> enter 140 in "marker size"
print at 100% scale ("actual size"), then verify with the ruler:



ars // field recipe – hand-draw marker id 0

No printer? id 0 is one bar. It cannot be drawn mirrored (its mirror is itself) and rotation does not matter (the detector resolves it). Only the GEOMETRY can fail, so use a straightedge. Any side length S works:

1. Draw a square, side S , and fill it solid black (dark + matte pen).
2. Leave (or paint) ONE white bar: width $S/7$,
starting $S/7$ in from the left edge,
running from $S/7$ below the top to $S/7$ above the bottom.
3. Keep a white margin of at least $S/7$ all around the square.
4. MEASURE your actual square and enter that size in the app.

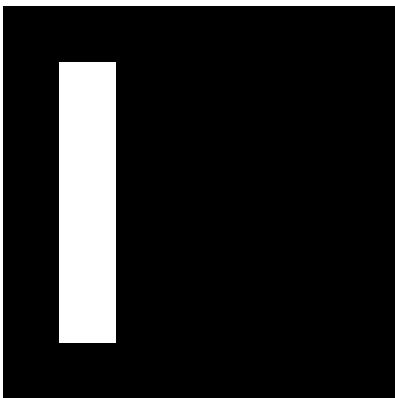
Tolerances: bar placement is forgiving (~a third of a cell); edges should be reasonably straight (wobble adds pose jitter; fusion smooths it).

Example: $S = 140$ mm $\rightarrow$ bar 20 mm wide, 20 mm from the left, 100 mm tall.



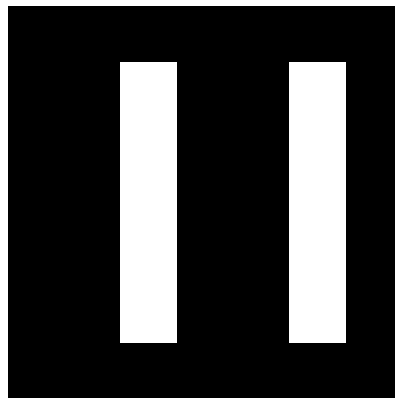
ars // field alphabet – the three drawable markers

The classic dictionary contains exactly THREE pure-stripe markers. One technique draws them all: fill a black square (side S), leave white vertical bars – every bar $5 \cdot S/7$ tall, $S/7$ clear at top and bottom. All three are mirror-proof (a mirrored drawing reads as a 180 deg rotation of the same id). Three markers on one sheet = a hand-drawn reference mat: a full constellation datum, no printer required.



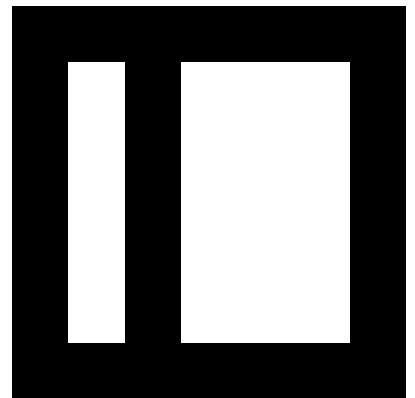
id 0

one thin bar at $1S/7$



id 682

thin bars at $2S/7$ and $5S/7$



id 341

thin at $1S/7$ + FAT ($3S/7$) at $3S/7$

keep white margins $\geq S/7$ around each square; measure each real S and enter it